

LOBEFIT Manual: 1.0 version

This document is available at:

<https://github.com/zhaoxi103/LOBEFIT>

Sep 10, 2024

Contents

1	Overview	3
2	User Requirements.....	3
2.1	System Requirements	3
2.2	License	5
3	Instructions	6
3.1	Installation.....	6
3.2	How to Run in CUI Mode	7
3.2.1	<i>Create Configuration File.....</i>	<i>7</i>
3.2.2	<i>Run in the console</i>	<i>8</i>
3.3	How to Run in GUI Mode.....	9
3.3.1	<i>Input and choose parameters</i>	<i>9</i>
3.3.2	<i>Run in GUI</i>	<i>11</i>
3.4	Result Files and Plot Fitting errors with LOBEFIT	13
3.4.1	<i>Result Files</i>	<i>13</i>
3.4.2	<i>Plot Fitting errors with GUI.....</i>	<i>14</i>
4	How to integrate your own model (Secondary development).....	15
Appendix A	File Formats	17
A.1	Input LEO satellite precise orbit File (SP3 format)	17
A.2	Solution File: eph_para	19
A.3	Fitting Error File: URE_file.....	20
A.4	Configuration File	21
Appendix B	API References	22
Appendix C	Models and Algorithms	24
C.1	Time and Coordinates System.....	24
C.2	Basic 16-parameter broadcast ephemeris model.....	25

C.3	17~22-parameter broadcast ephemeris model within the software	26
C.4	Fitting errors and UREs	29
References		30

1 Overview

LOBEFIT (LeO satellite Broadcast Ephemeris FITting) is an open-source software integrated the automatic differentiation technique to perform ephemeris parameter fitting for different models of LEO satellites. It was entirely written in Python and could be efficiently executed in both CUI and GUI modes. The application of automatic differentiation effectively eliminated the need for developers to manually compute complex partial derivatives within the ephemeris models, substantially facilitating expansion and iterative refinement for users.

The features of LOBEFIT are:

- (1) Performing the fitting of LEO satellite broadcast ephemeris parameters
- (2) Obtaining LEO satellite broadcast ephemeris parameters fitting accuracy and visualizing
- (3) Calculating satellite position vectors with various broadcast ephemeris models
- (4) Calculating the partial derivatives of various broadcast ephemeris models using automatic differentiation technique with JAX library^[3]
- (5) Realizing the mutual conversion of different time systems

2 User Requirements

2.1 System Requirements

The software is written in Python. A Python version 3.9 or higher is recommended. The software was tested on Windows 10 (64bit) and CentOS 7.6 Linux. What's more, some additional Python libraries are needed. These libraries are JAX^[3], Numpy, NiceGUI^[4], et al..

On Windows 10 (64bit), the version of Python and the Python packages in our test are:

```
python==3.10.9
jax==0.4.31
matplotlib==3.7.0
nicegui==2.1.0
numpy==2.1.1
```

On CentOS 7.6, the version of Python and the Python packages in our test are:

```
python==3.9.10
```

```
jax==0.4.25  
nicegui==2.1.0  
matplotlib==3.1.2  
numpy==1.21.4
```

Notes: The other used Python libraries are Python standard libraries os, argparse, socket.

2.2 License

MIT License

Copyright (c) 2024 zhaoxi103

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

3 Instructions

3.1 Installation

Download or use “git clone” command to obtain the package from the website <https://github.com/zhaoxi103/LOBEFIT> and extract the program package to appropriate directory <install dir>. The LOBEFIT directory structure is as follows.

We recommend to use a virtual-environment (or Anaconda equivalent). Just use “pip install” to install Python libraries as shown in 2.1.

LOBEFIT-main

\src	: Source code of LOBEFIT
\orbddata	: Sample data for precise LEO satellite orbits in sp3 format ^[6]
\output	: Directory used for saving output Files
\doc	: Document files
config_new.ini	: Sample configuration file for input settings

3.2 How to Run in CUI Mode

3.2.1 Create Configuration File

One should create a file of txt format for configuration at any path. For an example, one can create a file named "config.ini" at "/home/zhaox/code".

The configuration file specifies input parameters using the format "KEY = VALUE". The comprehensive definitions for each key and their corresponding values are presented in Table 1.

Table 1 The definitions of each key and their corresponding values in the configuration file

KEY	The definition of "KEY"	Example value
Start_Time	the first epoch for fitting (GPS Time)- year month day hour minute second	2020 7 28 10 0 0.00
Indicator	the abbreviated designation for LEO satellite within the input file specifying precise satellite orbits	K01
Input_File	The complete path and the file name of the input precise satellite orbit file	seb2000.sp3
Input_Type	the type of the input precise satellite orbit file 'PV' denotes the orbit file encompasses both the position and velocity data of the LEO satellite, whereas 'P' signifies that the file exclusively contains the position data of the LEO satellite.	PV
Interval	the interval between neighboring epochs (the unit is second)	10
Para_Num	the number of LEO satellite broadcast ephemeris parameters	18
Para_Case	the case number in the supplementary material pertaining to a specific scheme	case2
Fitting_Time	the interval between selected initial epoch and ending epoch utilized for the fitting process (the unit is second)	1200
URER	URE weight factors in the radial direction for different orbital altitude	0.4187
URETN	URE weight factors in the cross and along direction for different orbital altitude	0.6421
OutPath	the absolute path to the output files	/home/zhaox

EXAMPLE

[config]

Start_Time = 2023 5 20 0 00 0.000

Indicator = 202

Input_File = /home/zhaox/ephfit/code/orbdata/modified_grac1400.txt

Input_Type = PV

Interval = 1

Para_Num = 17

Para_Case = case1

Fitting_Time = 900

URER = 0.4557

URETN = 0.6294

OutPath = /home/zhaox/ephfit/code/output

3.2.2 Run in the console

Execute the LOBEFIT in CUI mode from the **src** directory.

```
python brdEphFit.py -ini /home/zhaox/ephfit/code/config.ini
```

Here we assume that the path of the configuration file that we created is /home/zhaox/ephfit/code/config.ini. Waiting for a moment, we can obtain the result.

3.3 How to Run in GUI Mode

We designed a simple UI using NiceGUI^[4], a Python-based UI framework that works smoothly with the web browsers. This allows the web browser to serve as the frontend for our Python code. We incorporated a web-based UI into the software design, which also addresses cross-platform compatibility challenges.

Execute the LOBEFIT in GUI mode from the **src** directory.

The interface can be initiated by entering the command "python NiceGUIApp.py" in the console. The application is subsequently displayed within the default web browser environment on the local host.

3.3.1 Input and choose parameters

The UI contains 4 tab panels, which are "1. LEO satellite Option", "2. Fitting Option", "3. Output Option" and "4. URE Error Visualization". Users should first input or choose parameters as you like in "1. LEO satellite Option", "2. Fitting Option" and "3. Output Option". "1. LEO satellite Option" contains the setting parameters about the specific LEO satellite, including "Input_File", "Input_Type", "Interval" and "Indicator"; "2. Fitting Option" contains the setting parameters about fitting, including "Start_Time", "Fitting_Time", "Para_Num" and "Para_Case"; "3. Output Option" contains the setting parameters about the output results, including "URER", "UREN" and "OutPath". These have the same meaning as the "KEY" shown in Table 1.

EXAMPLE

(1) Switch to "1. LEO satellite Option":

LEOEPHFIT

127.0.0.1:10000

1. LEO SATELLITE OPTION

2. FITTING OPTION

3. OUTPUT OPTION

4. URE ERROR VISUALIZATION

Input_File

D:\ephfit\orbddata\se3b2f

Input_Type

PV

Interval(s)

10

Indicator

K01

(2) Switch to "2. Fitting Option":

LEOEPHFIT

127.0.0.1:10000

1. LEO SATELLITE OPTION

2. FITTING OPTION

3. OUTPUT OPTION

4. URE ERROR VISUALIZATION

Start_Time

2019 7 18 23 00 00.000

Fitting_Interval(s)

1200

Para_Num

20

Para_Case

case2

(3) Switch to "3. Output Option":

- (4) Click the button “(A) OK” and the configuration file “config_nicegui.ini” will be generated according to your setting options in the parent directory of ‘src’. After Clicking the button “(A) OK”, a notification “The config file has been generated!” will be displayed on the top of the screen. The content of “config_nicegui.ini” is shown in the following Figure.

```
[config]
Start_Time   = 2019 7 18 23 00 00.000
Indicator    = K01
Input_File   = D:\ephfit\orbdata\se3b2000.sp3
Input_Type   = PV
Interval     = 10
Para_Num     = 20
Para_Case    = case2
Fitting_Time = 1200
URER         = 0.4557
URETN        = 0.6294
OutPath      = D:\ephfit\output
```

3.3.2 Run in GUI

After setting options according to 3.3.1, click the button “(B) CALCULATE”. The fitting process will be then triggered.

1. LEO SATELLITE OPTION 2. FITTING OPTION 3. OUTPUT OPTION 4. URE ERROR VISUALIZATION

URER
0.4557

URETN
0.6294

OutPath
D:\ephfit\output

(A) OK (B) CALCULATE

Then a notification “Computing...” will be displayed on the top of the screen until the fitting process finished.



After the fitting process finished, a notification “Done” will be displayed on the top of the screen.



3.4 Result Files and Plot Fitting errors with LOBEFIT

3.4.1 Result Files

The result files are in the directory which we set in “OutPath”.

```
[config]
Start_Time   = 2019 7 18 23 00 00.000
Indicator    = K01
Input_File   = D:\ephfit\orbdata\se3b2000.sp3
Input_Type   = PV
Interval     = 10
Para_Num     = 20
Para_Case    = case2
Fitting_Time = 1200
URER         = 0.4557
URETN        = 0.6294
OutPath      = D:\ephfit\output
```

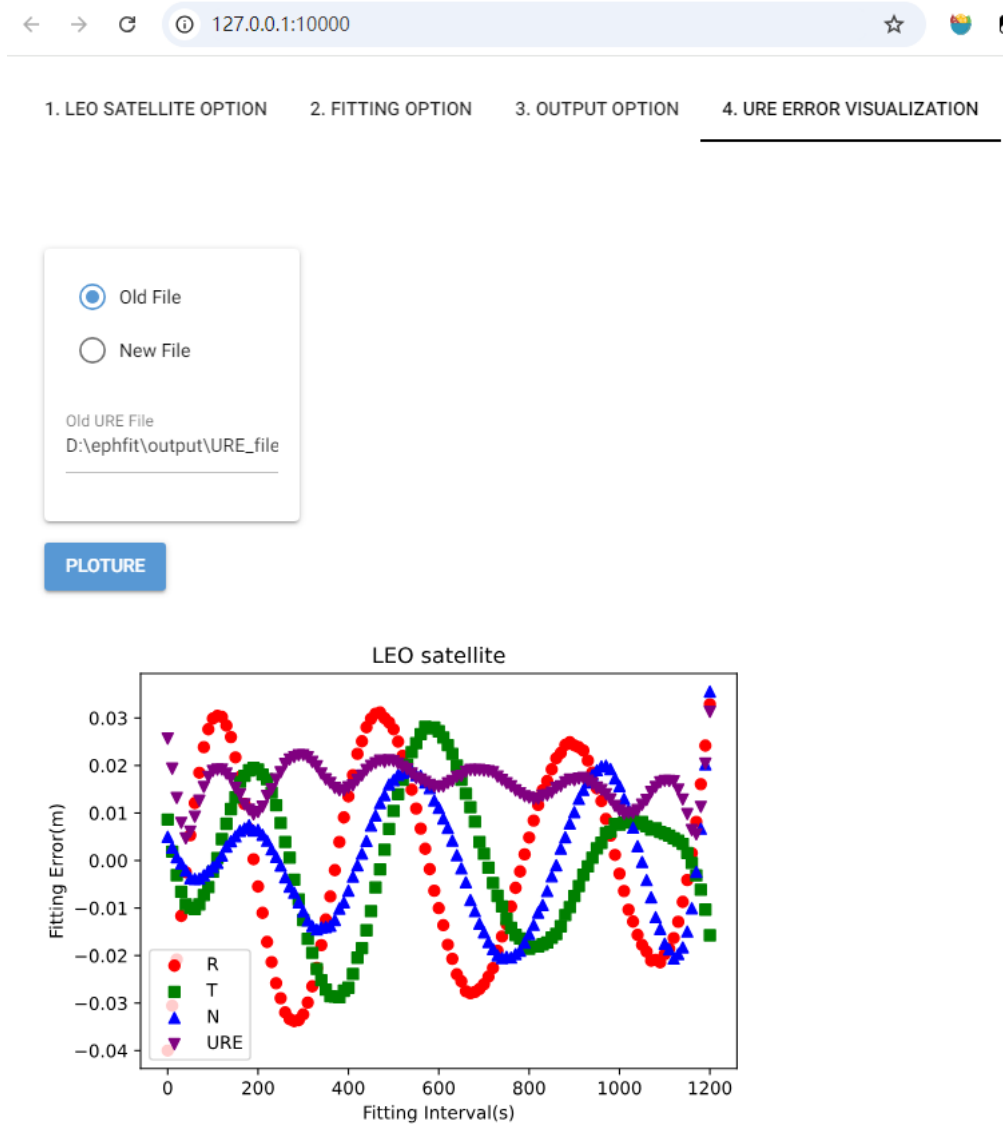
The screenshot shows a web browser window with the address bar displaying "127.0.0.1:10000". The page has four tabs: "1. LEO SATELLITE OPTION", "2. FITTING OPTION", "3. OUTPUT OPTION", and "4. URE ERROR VISUALIZATION". The "3. OUTPUT OPTION" tab is selected. Below the tabs, there is a form with three input fields: "URER" with the value "0.4557", "URETN" with the value "0.6294", and "OutPath" with the value "D:\ephfit\output". The "OutPath" field is highlighted with an orange border. At the bottom of the form, there are two buttons: "(A) OK" and "(B) CALCULATE".

If you finish the fitting process successfully, you will obtain two text files “eph_para” and “URE_file”. “eph_para” contains the estimated broadcast ephemeris parameters and “URE_file” contains the fitting errors between broadcast ephemeris and the precise orbits. For the contents of these two files, refer Appendix A.2 and A.3.

3.4.2 Plot Fitting errors with GUI

In GUI mode, the function of “Plot Fitting Errors” is provided. There two options we can choose. “Old File” means that if you have saved previous fitting error result and would like to plot, just choose “Old File” and input the absolute path of the fitting error result file. To choose “New File” means that the program will plot the “URE_file” saved in “OutPath” that you set and do not input manually the absolute path of the fitting error result file, but you have to ensure that you have input your option at the tab panel “1. LEO satellite Option”, “2. Fitting Option”, “3. Output Option” and click “(A) OK” to generate the configuration file and then perform the fitting process successfully.

Then you can click the button “PLOTURE”, the figure will be displayed on the screen.



4 How to integrate your own model (Secondary development)

The software can integrate GNSS satellite broadcast ephemeris models and your own LEO satellite broadcast ephemeris models. The most critical aspect is to make modifications to the two functions.

(1) Add your own model in the python file “brdModel.py” like:

```
def your_own_model(x, tk, toe):
    ...
    ...(your own specific implementation)
    ...
    return jnp.asarray([pos_x, pos_y, pos_z])
```

Here x represents the broadcast ephemeris parameters, tk means all of the chosen fitting epochs, toe is the ephemeris reference time, which is a fixed parameter. pos_x, pos_y and pos_z mean the satellite position in ECEF coordinate system.

(2) Add the “import” and “if condition” for your own model in the function “formMatrixSingleEpoch” in the python file “brdParDeriv.py” like:

```
from brdModel import your_own_model
...
...
...
elif (the self-defined condition for your own model):
    x_cal_pos = your_own_model(x_brd, tk, toe)
    f = jax.jacrev(your_own_model)
    dfdx = f(x_brd, tk, toe)
...
...
```

Here x_brd represents the broadcast ephemeris parameters, tk means all of the chosen fitting epochs, toe is the ephemeris reference time, which is a fixed parameter. x_cal_pos mean the satellite position in ECEF coordinate system.

(3) Modify other functions about reading configuration and UI.

Appendix A File Formats

A.1 Input LEO satellite precise orbit File (SP3 format)

The precise orbit file gives the orbit information of LEO satellites at certain time intervals, including satellite coordinates, satellite clock errors and optionally motion velocity, coordinate standard deviation and satellite clock errors standard deviation. The current standard format is SP3 (Standard Product 3), including SP3-a, SP3-c and SP3-d. A brief description of SP3-d is given below, and the specific file format can be found at [sp3d.pdf^{\[6\]}](#).

DESCRIPTION

(1) File header

- The first line starts with “#a”, “#c” or “#d”, indicating the SP3 file type; the third character is “P” or “V”, marking the position or speed; the penultimate data is the reference frame.
- The second line, beginning with “##”, records the GPS week for the corresponding day, the seconds of week, the sampling interval, the integer part and the fractional part of the MJD.

(2) Data blocks

- The line starting with “*” marks the beginning of the epoch, followed by the corresponding epoch time;
- The first character of the data line is "P" or "V", followed by the PRN of the satellite;
- If the first letter is "P", the XYZ coordinates and clock deviation of the satellite will be recorded;
- If the first letter is "V", the change rate of satellite XYZ velocity and clock deviation is recorded;
- The end-of-file marker is "EOF".

EXAMPLE

```

1  #aP2023 5 20 22 0 0.00000000 9361 U+u IGS14 FIT ESA*
2  ## 2262 597600.00000000 10.00000000 60084 79200.0000
3  + 1 K01 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
4  + 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
5  + 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
6  + 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
7  + 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
8  ++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
9  ++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
10 ++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
11 ++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
12 ++ 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
13 %c L SENTINEL-3B
14 %c cc cc ccc ccc cccc cccc cccc cccc ccccc ccccc ccccc ccccc
15 * 2023 5 20 22 0 0.00000000
16 PK01 6978.732207 278.671215 -1688.493348 0.000000 0 0
17 VK01 1.787523 -1.588397 7.154569 0.000000
18 * 2023 5 20 22 0 10.00000000
19 PK01 6996.222124 262.759775 -1616.857916 0.000000 0 0
20 VK01 1.710427 -1.593844 7.172388 0.000000
21 * 2023 5 20 22 0 20.00000000
22 PK01 7012.940104 246.795271 -1545.048153 0.000000 0 0
23 VK01 1.633137 -1.599009 7.189435 0.000000
24 * 2023 5 20 22 0 30.00000000
25 PK01 7028.884240 230.780537 -1473.071793 0.000000 0 0
26 VK01 1.555660 -1.603890 7.205708 0.000000

```

Example of a SP3-d file

A.2 Solution File: eph_para

DESCRIPTION

This file contains LEO satellite broadcast ephemeris parameters and corresponding meaning for each parameter.

EXAMPLE

```

1 The numbers of parameters used for this resolution is: 20
2 toe: reference time ephemeris 4.296000000000e+05
3 sqra: square root of semi-major axis 2.679935966712e+03
4 dtv: eccentricity 1.198057883911e-03
5 inc0: inclination angle at reference time 1.721489703318e+00
6 Omega0: longitude of ascending node of orbit plane at weekly epoch -4.374007841867e-01
7 w: argument of perigee 1.588387853641e+00
8 ma: mean anomaly at reference time 1.628341460766e-01
9 dn: mean motion correction term -2.581832725127e-07
10 idot: rate of inclination angle -1.919512970280e-08
11 Omegadot: rate of right ascension 2.046887692616e-07
12 crc: amplitude of the cosine harmonic correction term to the orbit radius 1.562224501794e+03
13 crs: amplitude of the sine harmonic correction term to the orbit radius -3.442925043768e+01
14 cuc: amplitude of the cosine harmonic correction term to the argument of latitude -5.226333452116e-06
15 cus: amplitude of the sine harmonic correction term to the argument of latitude 9.938764215579e-05
16 cic: amplitude of the cosine harmonic correction term to the angle of inclination -4.995243056629e-06
17 cis: amplitude of the sine harmonic correction term to the angle of inclination 1.346261556889e-07
18 crc3: newly introduced parameter -2.318089012155e+01
19 crs3: newly introduced parameter -4.634195606146e+00
20 Adot: newly introduced parameter -1.297243199089e-01
21 Adotdot: newly introduced parameter -4.638319140792e-05

```

A.3 Fitting Error File: URE_file

DESCRIPTION

This file contains the fitting errors between the estimated broadcast ephemeris and the input precise orbits.

- Column 1: Modified Julian days
- Column 2: GPS week Seconds
- Column 3-6: the fitting errors in the radial, along-track and cross directions, as well as the User Range Errors (UREs)

EXAMPLE

1	58682.000000	428400.000000	-0.040003	0.008622	0.004978	0.025675
2	58682.000000	428410.000000	-0.030548	0.001887	0.002739	0.019323
3	58682.000000	428420.000000	-0.020831	-0.003077	0.000715	0.013194
4	58682.000000	428430.000000	-0.011654	-0.006569	-0.000624	0.007932
5	58682.000000	428440.000000	-0.002468	-0.009231	-0.002101	0.004675
6	58682.000000	428450.000000	0.005322	-0.009967	-0.003708	0.006107
7	58682.000000	428460.000000	0.012091	-0.010194	-0.003875	0.009243
8	58682.000000	428470.000000	0.018470	-0.009216	-0.003781	0.012588
9	58682.000000	428480.000000	0.023853	-0.007246	-0.003124	0.015497
10	58682.000000	428490.000000	0.027630	-0.005599	-0.002180	0.017630
11	58682.000000	428500.000000	0.029867	-0.002352	-0.001371	0.018848
12	58682.000000	428510.000000	0.030448	0.000551	-0.000504	0.019169
13	58682.000000	428520.000000	0.030229	0.004510	0.001064	0.019149
14	58682.000000	428530.000000	0.028395	0.007733	0.002969	0.018311
15	58682.000000	428540.000000	0.025987	0.010820	0.004079	0.017275
16	58682.000000	428550.000000	0.021703	0.013280	0.005096	0.015281
17	58682.000000	428560.000000	0.017579	0.016577	0.006270	0.013966
18	58682.000000	428570.000000	0.011929	0.017834	0.006735	0.011848

A.4 Configuration File

DESCRIPTION

A configuration file is a text file which contains the **Keyword = Value** form records indicating the various options. For the contents of the configuration file, refer 3.2.1.

EXAMPLE

```
[config]
Start_Time   = 2019 7 18 23 00 00.000
Indicator    = K01
Input_File   = D:\ephfit\orbddata\se3b2000.sp3
Input_Type   = PV
Interval     = 10
Para_Num     = 20
Para_Case    = case2
Fitting_Time = 1200
URER         = 0.4557
URETN        = 0.6294
OutPath      = D:\ephfit\output
```

Appendix B API References

The following table shows the list of RTKLIB library functions.

Table B-1 LOBEFIT API function List

Function	Description	Source Program
Constant values		
MU_GPS	Earth gravitational constant	brdConstant.py
OMGE	earth angular velocity (IS-GPS) (rad/s)	brdConstant.py
nMaxIter	The maximum number for iteration when fitting	brdConstant.py
Time functions		
doy_ymd()	converting doy to ymd or ymd to doy	time_convert.py
doy_mjd()	converting doy to mjd or mjd to doy	time_convert.py
mjd_gpswk()	converting the Mjd to GPS week or GPS week to Mjd	time_convert.py
sod_hms()	converting Sod(seconds of day) to Hms(hour minute second) or Hms to Sod	time_convert.py
sow_wkd()	converting Sow(second of week) to Wkd(week day) or Wkd to Sow	time_convert.py
dmjd_ymdhms()	converting Dmjd(mjd.sod) to Ymdhms(year, mon, doy, hh, mm, ss) or Ymdhms to Dmjd	time_convert.py
ymdhms_dif()	calculating the time difference between two time	time_convert.py
ymdhms_inc()	calculating the time after increase	time_convert.py
calGMST()	calculating GMST	time_convert.py
calGAST()	calculating GAST	time_convert.py
Input/output functions		
GetIniKeyString()	get the input parameter from a single line	brdEphFit.py
readConfig()	retrieves input parameters from the configuration file	brdEphFit.py
readsp3File()	read satellite orbit data to the array	brdEphFit.py
outputResult()	Output the estimated parameters and fitting errors	brdEphFit.py
Broadcast ephemeris models and automatic differentiation functions		
calSatPos_16para() ~ calSatPos_22para_case3()	calculate satellite position vectors using 16~22 parameter broadcast ephemeris models	brdModel.py
formMatrixSingleEpoch()	calculate the partial derivatives of satellite position vectors w.r.t. broadcast ephemeris parameters at a single epoch	brdParDeriv.py

Function	Description	Source Program
formMatrixManyEpochs()	accumulate the partial derivatives of satellite position vectors w.r.t. broadcast ephemeris parameters at many epochs	brdParDeriv.py
formMatrixManyEpochs_parallel()	accumulate parallelly the partial derivatives of satellite position vectors w.r.t. broadcast ephemeris parameters at many epochs, using Python ProcessPoolExecutor	brdParDeriv.py
Broadcast ephemeris fitting functions		
extract_orbit_and_time()	Extract the epochs and corresponding satellite orbits used for fitting	brdEphFit.py
initial_fiteph()	obtain initial broadcast ephemeris parameters before iteration for fitting	brdEphFit.py
ephfit()	perform complete broadcast ephemeris fitting process	brdEphFit.py
GUI functions		
brdfit_main_gui()	perform complete broadcast ephemeris fitting process for GUI	NiceGUIApp.py
mycalonClick()	callback function when click the “(B) CALCULATE” button	NiceGUIApp.py
plotURE()	Plot fitting errors on the screen	NiceGUIApp.py
find_free_port()	find a free port for web-based GUI	NiceGUIApp.py

Notes: The time system and corresponding concept in time_convert.py could refer C.1 and SOFA^[5].

Appendix C Models and Algorithms

This appendix briefly describes the models and algorithms involved in LOBEFIT.

C.1 Time and Coordinates System

LOBEFIT internally uses GPST (GPS Time) for data handling and fitting algorithms. One of the reasons why using GPST is to avoid leap seconds handling. The start epoch of GPST is 00:00:00 UTC on January 6, 1980. Moreover, GPST often can be expressed as a GPS week number and TOW (time of week) in seconds, Year and day of year, etc. The specific concepts and definitions can be found from SOFA^[5].

The satellite positions in LOBEFIT are internally represented as the X, Y, Z components in an ECEF (earth center earth fixed) coordinates system. What ECEF frame used is not explicitly defined but depends on the satellite ephemeris and the predefined base station position. For example, with GPS signals and navigation data, the single point positioning results are obtained in WGS84.

C.2 Basic 16-parameter broadcast ephemeris model

Table C-1 Basic 16-parameter broadcast ephemeris model^[1]

calculation steps	Formula
1. Semi-Major Axis	$A = (\sqrt{A})^2$
2. Computed Mean Motion (rad/sec)	$n_0 = \frac{\sqrt{\mu}}{\sqrt{A^3}} \quad \mu = 3.986005 \times 10^{14} \text{m}^3/\text{s}^2$
3. Time from ephemeris reference time	$t_k = t - t_{oe}$
4. Corrected Mean Motion	$n = n_0 + \Delta n$
5. Mean Anomaly	$M_k = M_0 + nt_k$
6. Kepler's equation may be solved for Eccentric Anomaly (E_k) by iteration	$M_k = E_k - e \sin E_k$
7. True Anomaly	$v_k = \arctan\left(\frac{\sqrt{1-e^2} \sin E_k}{\cos E_k - e}\right)$
8. Argument of Latitude	$\Phi_k = v_k + \omega$
9. Second Harmonic Perturbations	$\begin{cases} \delta u_k = C_{us} \sin 2\Phi_k + C_{uc} \cos 2\Phi_k \\ \delta r_k = C_{rs} \sin 2\Phi_k + C_{rc} \cos 2\Phi_k \\ \delta i_k = C_{is} \sin 2\Phi_k + C_{ic} \cos 2\Phi_k \end{cases}$
10. Corrected Argument of Latitude	$u_k = \Phi_k + \delta u_k$
11. Corrected Radius	$r_k = A(1 - e \cos E_k) + \delta r_k$
12. Corrected Inclination	$i_k = i_0 + \delta i_k + idot \cdot t_k$
13. Positions in orbital plane	$\begin{cases} x_k = r_k \cos u_k \\ y_k = r_k \sin u_k \end{cases}$
14. Corrected Longitude of Ascending Node	$\Omega_k = \Omega_0 + (\dot{\Omega} - \omega_e)t_k - \omega_e t_{oe} \quad \omega_e = 7.29211515 \times 10^{-5} \text{rad/s}$
15. Earth-fixed coordinates of LEO satellites	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_k \cos \Omega_k - y_k \cos i_k \sin \Omega_k \\ x_k \sin \Omega_k + y_k \cos i_k \cos \Omega_k \\ y_k \sin i_k \end{bmatrix}$

C.3 17~22-parameter broadcast ephemeris model within the software

The 17-22-parameter broadcast ephemeris model is characterized by the incorporation of supplementary formulas that correspond to the additional parameters.

Table C-2 The additional parameters and formulas of different LEO ephemeris parameter set

Parameter No.	Case	Additional parameters	Additional formulas
16	--	--	--
17	Case1	$\dot{A}$	$A_k = (\sqrt{A})^2 + \dot{A} \cdot t_k$
	Case2	$\dot{n}$	$n = n_0 + \Delta n + \dot{n} \cdot t_k$
	Case3	$\dot{r}$	$r_k = A(1 - e \cos E_k) + \delta r_k + \dot{r} \cdot t_k$
18	Case1	C_{rs3}, C_{rc3}	$\delta r_k = C_{rs} \sin 2\Phi_k + C_{rc} \cos 2\Phi_k + C_{rs3} \sin 3\Phi_k + C_{rc3} \cos 3\Phi_k$
	Case2	C_{us3}, C_{rc3}	$\delta u_k = C_{us} \sin 2\Phi_k + C_{uc} \cos 2\Phi_k + C_{us3} \sin 3\Phi_k + C_{uc3} \cos 3\Phi_k$
	Case3	$\dot{n}, \ddot{n}$	$n = n_0 + \Delta n + \dot{n} \cdot t_k + \frac{1}{2} \ddot{n} \cdot t_k^2$
19	Case1	$\dot{A}, \ddot{A}, \dot{n}$	$A_k = (\sqrt{A})^2 + \dot{A} \cdot t_k + \frac{1}{2} \ddot{A} \cdot t_k^2$
	Case2	$\dot{A}, \dot{n}, \ddot{n}$	$n = n_0 + \Delta n + \dot{n} \cdot t_k + \frac{1}{2} \ddot{n} \cdot t_k^2$
	Case3	$\dot{r}, C_{rs1}, C_{rc1}$	$\delta r_k = C_{rs} \sin 2\Phi_k + C_{rc} \cos 2\Phi_k + C_{rs1} \sin \Phi_k + C_{rc1} \cos \Phi_k + \dot{r} \cdot t_k$
20	Case1	$C_{rs3}, C_{rc3}, C_{us1}, C_{uc1}$	$\delta r_k = C_{rs} \sin 2\Phi_k + C_{rc} \cos 2\Phi_k + C_{rs3} \sin 3\Phi_k + C_{rc3} \cos 3\Phi_k$
	Case2	$\dot{A}, \ddot{A}, C_{rc3}, C_{rs3}$	

		$\begin{aligned} \delta u_k &= C_{us} \sin 2\Phi_k \\ &+ C_{uc} \cos 2\Phi_k \\ &+ C_{us1} \sin \Phi_k \\ &+ C_{uc1} \cos \Phi_k \end{aligned}$
Case3	$\dot{r}, \dot{u}, C_{us3}, C_{uc3}$	$\begin{aligned} A_k &= (\sqrt{A})^2 + \dot{A} \cdot t_k + \frac{1}{2} \ddot{A} \cdot t_k^2 \\ \delta r_k &= C_{rs} \sin 2\Phi_k \\ &+ C_{rc} \cos 2\Phi_k \\ &+ \dot{r} \cdot t_k \end{aligned}$
		$\begin{aligned} \delta u_k &= C_{us} \sin 2\Phi_k \\ &+ C_{uc} \cos 2\Phi_k \\ &+ C_{us3} \sin 3\Phi_k \\ &+ C_{uc3} \cos 3\Phi_k + \dot{u} \cdot t_k \end{aligned}$
Case1	$\dot{r}, C_{rs1}, C_{rc1}, C_{us3}, C_{uc3}$	$\begin{aligned} \delta r_k &= C_{rs} \sin 2\Phi_k \\ &+ C_{rc} \cos 2\Phi_k \\ &+ C_{rs1} \sin \Phi_k \\ &+ C_{rc1} \cos \Phi_k \\ &+ \dot{r} \cdot t_k \end{aligned}$
Case2	$\dot{n}, C_{rs1}, C_{rc1}, C_{us3}, C_{uc3}$	$\begin{aligned} u_k &= C_{us} \sin 2\Phi_k \\ &+ C_{uc} \cos 2\Phi_k \\ &+ C_{us3} \sin 3\Phi_k \\ &+ C_{uc3} \cos 3\Phi_k \\ n &= n_0 + \Delta n + \dot{n} \cdot t_k \end{aligned}$
21		$\begin{aligned} \delta r_k &= C_{rs} \sin 2\Phi_k \\ &+ C_{rc} \cos 2\Phi_k \\ &+ C_{rs3} \sin 3\Phi_k \\ &+ C_{rc3} \cos 3\Phi_k \end{aligned}$
Case3	$\dot{n}, C_{rs3}, C_{rc3}, C_{us1}, C_{uc1}$	$\begin{aligned} \delta u_k &= C_{us} \sin 2\Phi_k \\ &+ C_{uc} \cos 2\Phi_k \\ &+ C_{us1} \sin \Phi_k \\ &+ C_{uc1} \cos \Phi_k \end{aligned}$

22

			$\delta r_k = C_{rs} \sin 2\Phi_k$
Case1	$\dot{r}, \dot{u}, C_{rs1}, C_{rc1}, C_{us3}, C_{uc3}$		$+ C_{rc} \cos 2\Phi_k$ $+ C_{rs1} \sin \Phi_k$ $+ C_{rc1} \cos \Phi_k$ $+ \dot{r} \cdot t_k$
Case2	$\dot{n}, \ddot{n}, C_{rs1}, C_{rc1}, C_{us3}, C_{uc3}$		$\delta u_k = C_{us} \sin 2\Phi_k$ $+ C_{uc} \cos 2\Phi_k$ $+ C_{us3} \sin 3\Phi_k$ $+ C_{uc3} \cos 3\Phi_k$ $+ \dot{u} \cdot t_k$
			$n = n_0 + \Delta n + \dot{n} \cdot t_k + \frac{1}{2} \ddot{n} \cdot t_k^2$
Case3	$\dot{n}, \ddot{n}, C_{rs3}, C_{rc3}, C_{us1}, C_{uc1}$		$\delta r_k = C_{rs} \sin 2\Phi_k$ $+ C_{rc} \cos 2\Phi_k$ $+ C_{rs3} \sin 3\Phi_k$ $+ C_{rc3} \cos 3\Phi_k$ $\delta u_k = C_{us} \sin 2\Phi_k$ $+ C_{uc} \cos 2\Phi_k$ $+ C_{us1} \sin \Phi_k$ $+ C_{uc1} \cos \Phi_k$

C.4 Fitting errors and UREs

The orbit errors Δr can be divided into the radial (ΔR), along-track (ΔA), and cross-track (ΔC) directions. The orbit-only contribution to the URE is defined as a weighted average of RMS errors $A = \text{RMS}(\Delta A)$, $C = \text{RMS}(\Delta C)$, and $R = \text{RMS}(\Delta R)$, as follows:

$$URE = \sqrt{w_R^2 R^2 + w_{A,C}^2 (A^2 + C^2)}$$

The weight factors w_R and $w_{A,C}$ are positively related to the orbital altitude. The values of the weight factors for LEO satellites range from 400 to 1400 km and are listed in Table C-3^[2].

Table C-3 User range error (URE) weight factors for different orbital altitude LEO satellites

Orbit Altitude (km)	w_R	w_A	w_C
300	0.3743	0.6557	0.6557
400	0.4187	0.6421	0.6421
500	0.4557	0.6294	0.6294
600	0.4875	0.6174	0.6174
700	0.5153	0.6060	0.6060
800	0.5400	0.5952	0.5952
900	0.5622	0.5848	0.5848
1000	0.5823	0.5749	0.5749
1100	0.6007	0.5653	0.5653
1200	0.6176	0.5562	0.5562
1300	0.6331	0.5473	0.5473
1400	0.6476	0.5388	0.5388
1500	0.6610	0.5306	0.5306

References

- [1] IS-GPS-200(Revision N). Navstar GPS Space Segment/Navigation User Interfaces, August 22th, 2022.
- [2] Fang SC, Du L, Gao YP, et al. Orbital elements ephemerides and interfaces design of LEO satellites. *Acta Geod Cartogr Sin* 2019; 48(2): 198-206.
- [3] Bradbury J, Frostig R, Hawkins P, et al. JAX: composable transformations of Python+NumPy programs. 2018. <http://github.com/google/jax>
- [4] Schindler F, Trappe R. NiceGUI: Web-based user interfaces with Python. The nice way. (Version 2.1.0). 2024. <https://doi.org/10.5281/zenodo.13623828>
- [5] <http://www.iausofa.org/cookbooks.html>
- [6] <https://files.igs.org/pub/data/format/sp3d.pdf>